

Classification of Political Tweets

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Abstract: Text classification via machine learning has been used to perform various text categorization tasks such as sentiment analysis, topic categorization, and question classification. In this paper, we will use machine learning techniques to separate Tweets from members of U.S. House of Representatives by political affiliation. We will build a variety of models and discuss their technical contributions, similarities, and advantages. Finally, the performance of the different models is quantitatively analyzed and the future research direction is discussed.

1. Introduction and background:

Twitter is a public online forum where many voices discuss, debate, and share their views. It plays a prominent role in how politicians, media outlets, and advocacy organizations promote their agendas and engage with political issues. More and more politicians are turning to Twitter to reach larger audiences, and to monitor political opinions in real-time. Sometimes it is easy to tell apart Twitter users with left- or right-leaning political alignment by the account names, bios, or profile pictures. However, only a small fraction of users report their political affiliations. It is challenging to tell users' political party affiliations without obvious labels and titles.

One of Twitter's interesting features is that each tweet is limited to 280 characters. Thus, users have to express their opinions more directly, and they even develop metadata annotation schemes to compress substantial amounts of information into a tiny space; 'Hashtags'. For example, #DACA denotes that the tweet is about 'Deferred Action for Childhood Arrivals'.

The data set we chose to use is Democrat Vs. Republican Tweets from Kaggle.com. It is comprised of the latest 200 tweets as of May 2018 from each of the 433 Representatives who have a Twitter account (over 80,000 tweets total), along with their Twitter user name and political party. The U.S. House of Representatives, is one

of two chambers of the U.S. Legislature. Each state has a certain number of elected Representatives determined by the population of the state. California has the most Representatives, with 53, while 7 states (Alaska, Delaware, Montana, North Dakota, South Dakota, Vermont, and Wyoming) only have 1 Representative each.

2. Method

Text data is an extremely rich source of information, but performing statistical analysis on it can be challenging due to its unstructured nature. The subfield of data science interested in solving these problems is broadly known as Natural Language Processing (NLP). Text classification is just one of many tasks of NLP. It is the process of assigning tags or labels to text data. Popular choices of text classification algorithms include k-nearest neighbors, naive Bayes, logistic regression, decision trees, and random forests.

In this paper, we attempt various machine learning based methods which share the following basic structure: First, some hand-crafted features are extracted from the tweets, such as a count of the occurrence of words in the tweet. This is commonly referred to as “bag-of-words.” Next, those features are run through a classifier to predict which group the tweet belongs to (either Democrat or Republican).

The features will be extracted using a method known as term frequency–inverse document frequency (TFIDF), which quantifies how important each term is to each label by measuring how often it appears in documents assigned to that label, and then offsets that measurement by how often the term appears in *all* documents. This helps adjust for the fact that some terms appear in all documents, regardless of label, and are therefore bad predictors of group membership. The strongest predictors appear very often in documents belonging to one label, and very infrequently in documents belonging to other labels.

Framework for Text Classification:

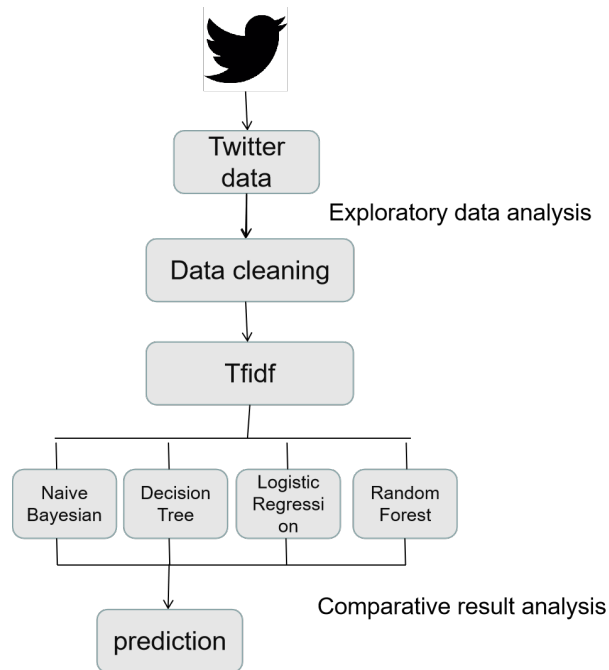


Figure 1. Framework for Text Classification

After feature extraction with TFIDF, we will use various machine learning models as classifiers. The scikit library in python will provide us with different models and algorithms to use as classifiers. The matplotlib and seaborn libraries will help us visualize the data and results. The pandas library will help us read CSV files. Finally, the NLTK library will help us remove unnecessary data from the dataset.

The following figure defines the process of text classification:

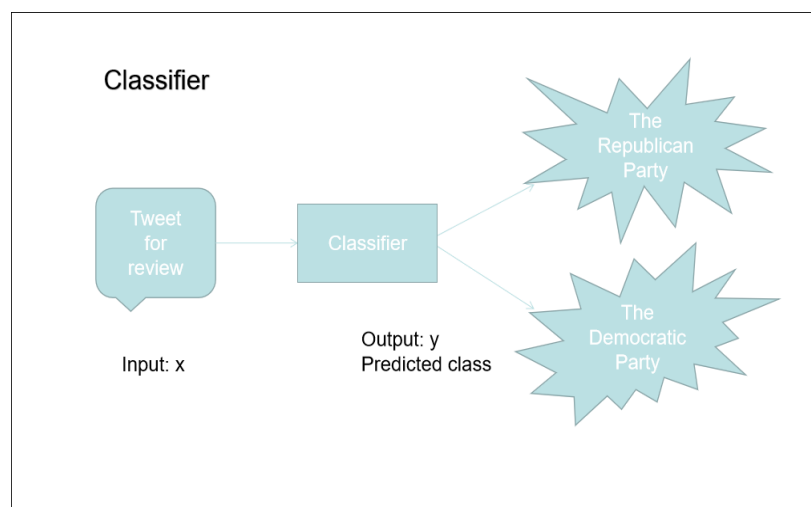


Figure 2. process of text classification

3. Data Cleaning

Because the original text may have numbers, stop words, and special symbols, we need to do text preprocessing to clean them. We used a word tokenizer provided by the NLTK library to tokenize a sentence, then we deleted stop words, numbers, emojis, URLs, punctuation, and special symbols from the tweet. We also converted all words of the tweet to lower case so that a word would be considered the same word whether it appeared at the beginning of a sentence or in the middle of a sentence.

4. Exploratory Data Analysis (EDA)

First, let's look at the number of members of different the parties represented in the data set.

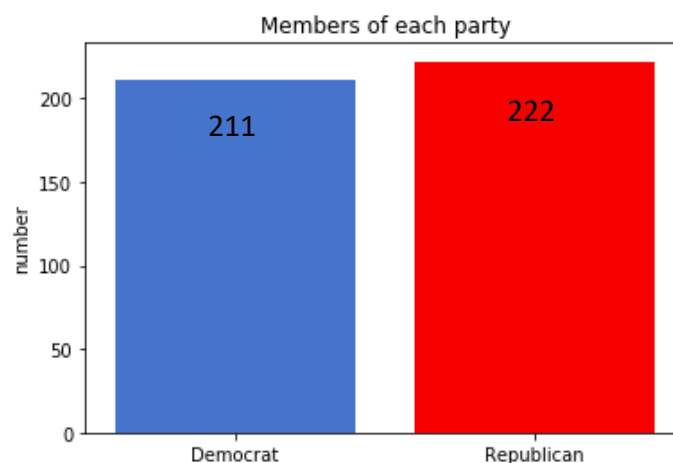


Figure 3. Members of each party

The House of Representatives has 435 total members. As of May 2020, it consists of 233 Democrats, 196 Republicans, and 1 Libertarian, but at the time this data was collected from Twitter, those numbers were slightly different. There are 222 Republican Representatives and 211 Democratic Representatives in the data set. 2 members of the House are not accounted for, and presumably don't use Twitter at all. In total, the data set contains 44392 tweets from Republicans and 42068 tweets from Democrats.

Next we investigated if any Representatives were on Twitter but used it very sparingly. The data set collected the 200 most recent tweets of each user, so if any user had a lot less than 200 tweets in the data set, it is safe to assume they barely used Twitter, so the data set collected every tweet they've ever done. The vast majority of Representatives had the full 200 tweets in the data set. A few had just under 200, but only one, Collin Peterson (D, Minnesota), had significantly less than 200 tweets. He only had 80 tweets.

We also analyzed and compared the average length of each tweet for the different parties.

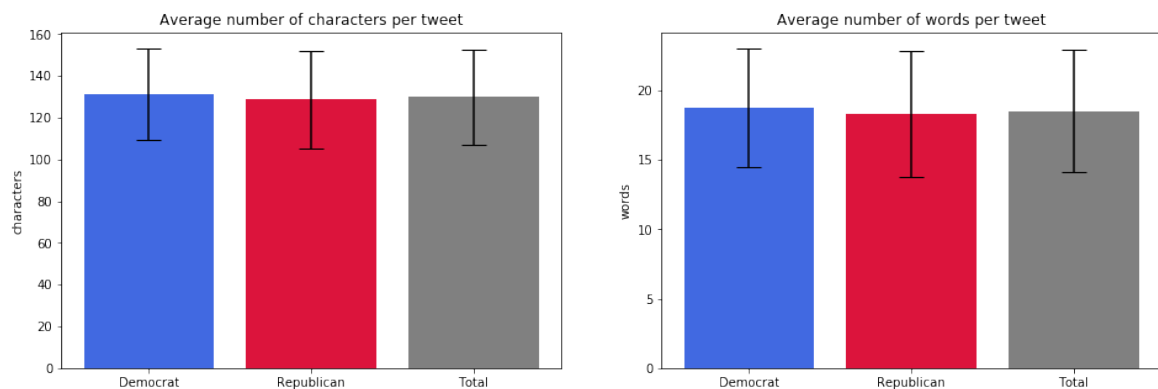


Figure 4. the average length of each tweet

The two parties are fairly similar, when it comes to average length of each tweet, both in word count and in character count. But the Democrat's tweets are just a tiny bit wordier on average than the Republican's tweets.

Next, we were interested in the actual composition of the tweets; What words appeared in them and how frequently. These word clouds represent the most common words (not counting stop words), that appear in Democratic Representative's tweets (left) and Republican Representative's tweets (right). The larger the word is, the more frequently it appeared.

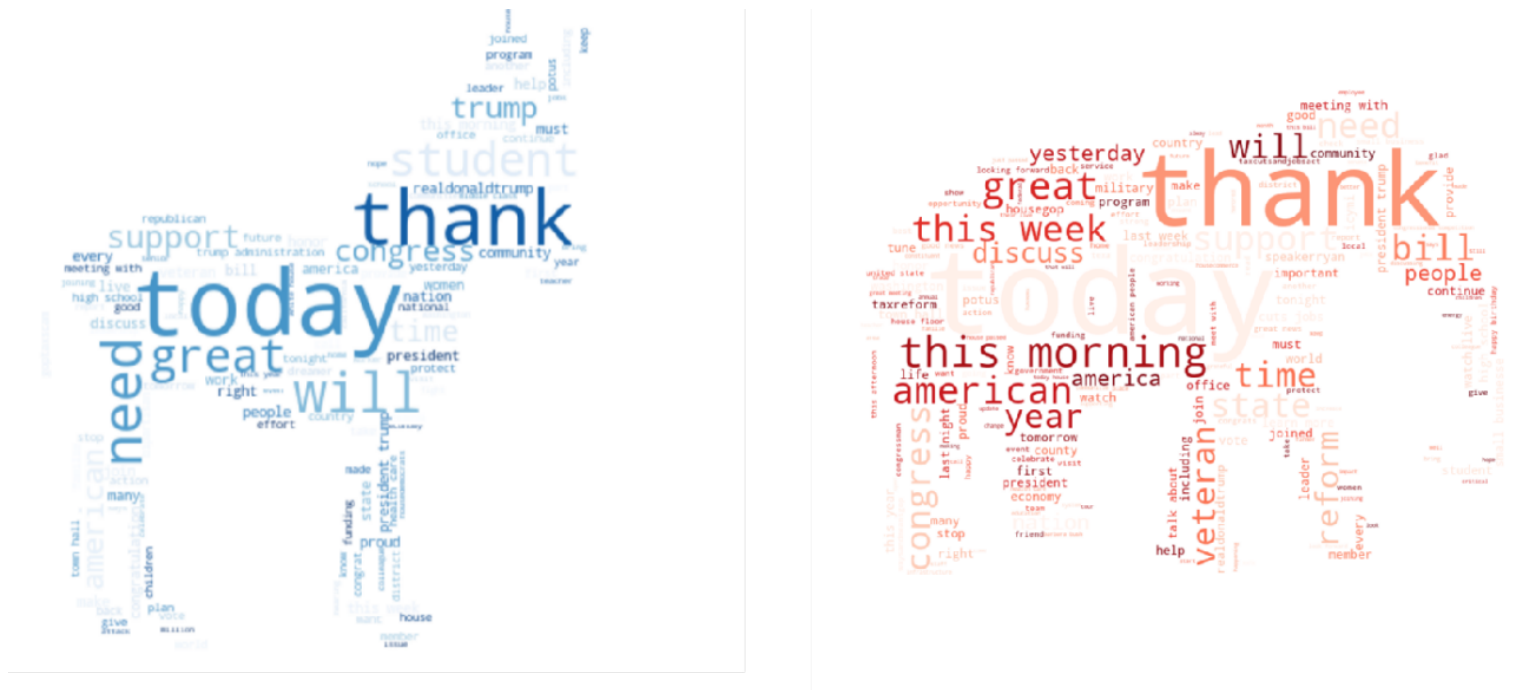


Figure 5. Word clouds

We see that some of the most common words appear in both parties' tweets, such as 'thank', 'today', 'American', 'will', 'support', etc. Hopefully our models will be able to ignore these commonly shared words and pick up on the words that will differentiate the parties. For example, I notice the word 'veteran' in the Republican's word cloud, but not in the Democrat's, and I notice the word 'women' in the Democrat's word cloud but not the Republican's.

Then, we analyzed the most commonly used hashtags by each party in their tweets.

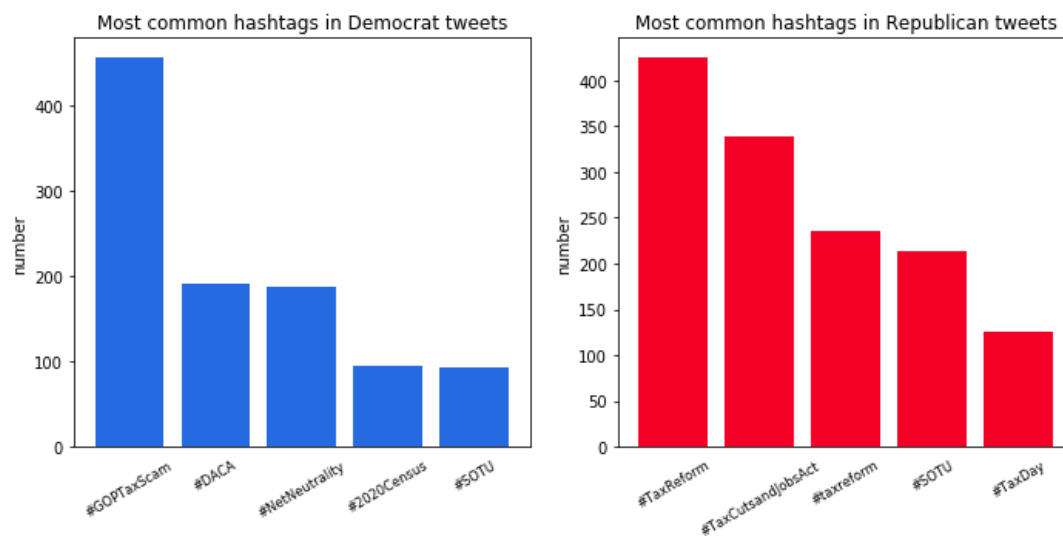


Figure 6. Most frequently used hashtags

Commonly used hashtags could be an insight into the topics that the Representatives care about and topics they think their constituents care about. We see that the only topic the two parties have in common in their top 5 most commonly used hashtags is #SOTU, which is in reference to the President's State of the Union speech. The other 4 out of 5 of the Republican's most common hashtags are all about taxes, which is pretty revealing about what they consider to be the most pressing issue. The Democrats, by contrast, seem to be interested in a more wide variety of issues such as Net Neutrality, DACA, and the 2020 Census.

5. TFIDF

The bag-of-words model is a simplifying representation used in NLP. In this model, each tweet is considered a separate document, and is represented as a collection of words by disregarding grammatical errors and even word order. We used sklearn's `TfidfVectorizer()` to compute the TFIDF matrix, which has one row for each tweet and one column for each unique word that appears across all tweets (ignoring stop words, punctuation, etc).

6. Model

We followed 7 steps in building our text classification model. The steps are as follows:

Step 1: Split Dataset.

We randomly selected 80% of the data as training data and 20% as test data.

Step 2: Define four machine learning models.

There are lots of supervised learning algorithm like naive Bayes, logistic regression, k-nearest neighbors, decision trees and so on. The scikit-learn library has implemented them for us so we can implement these algorithms with relatively little code. In our analysis, we used MultinomialNB DecisionTree, LogisticRegression and RandomForest.

Step 3: Train the model and make predictions.

We get the TFIDF matrix representation of the text, input the TFIDF matrix into the models defined above, conduct training, and then test with the test data to get the predicted labels from the different models.

Step 4: F1 score, precision, and recall values of all models.

F1 score, precision, and recall are three evaluation index for the binary classifiers. Precision means “given a class prediction from the classifier, how likely is it to be correct?” Recall means “given a class, will the classifier detect it?” F1 score considers both precision and recall. The calculation of the three indexes are as follows:

$$(1) \text{ F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

$$(2) \text{ Precision} = \frac{TP}{TP + FP}$$

$$(3) \text{ Recall} = \frac{TP}{TP + FN}$$

The term true positive (TP) denotes correctly identified prediction, true negative (TN) denotes correctly rejected prediction, false positive (FP) denotes incorrectly identified, and false negative (FN) denotes incorrectly rejected.

According to the labels obtained from different models, F1 score, precision and recall values were calculated and plotted into a histogram for comparison.

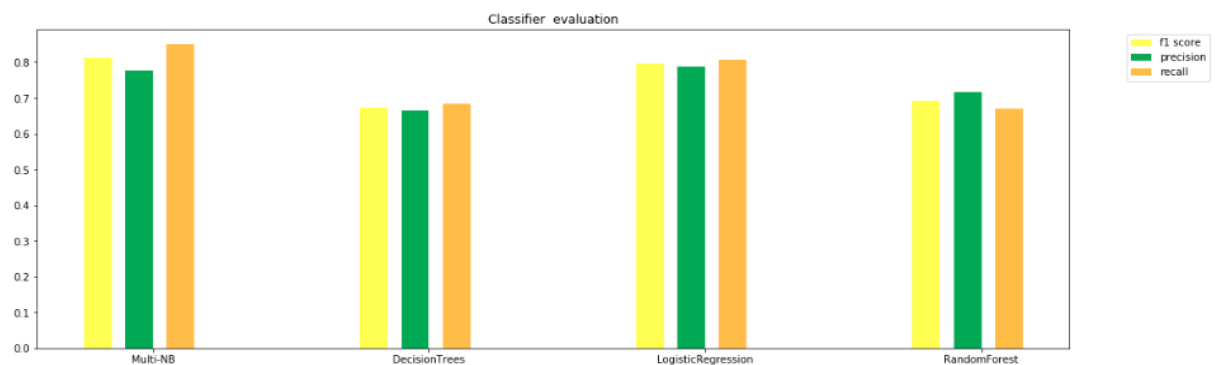


Figure 7. Classifier evaluation

As can be seen from the figure, the model with the best effect is multinomial naive Bayes, with an F1 score of about 0.812, while Decision Tree has the worst performance, with an F1 score of about 0.67.

Step5: Input a test case and output the predicted results of each model.

Mostly out of curiosity, and partly to test how the models perform, we worked out the projected results of the different models by entering a test sentence:

"I think we should lower taxes."

```
classifiers: model_NB, pred: Republican
classifiers: model_DT, pred: Democratic
classifiers: model_LR, pred: Republican
classifiers: model_RF, pred: Democratic
```

Both the multinomial naive Bayes and Logistic Regression model predicted this sentence as being tweeted by a member of the Republican party, while the other two considered it as having come from the Democratic Party. However, since as we know from the above analysis of the data that the Republicans are more focused on the topic of taxes, this sentence is more likely to belong to the Republicans.

Step 6: Draw the confusion matrix.

Since multinomial naive Bayes has the best results, we constructed the confusion matrix based on this model, and observed the performance of the model by analyzing the confusion matrix.

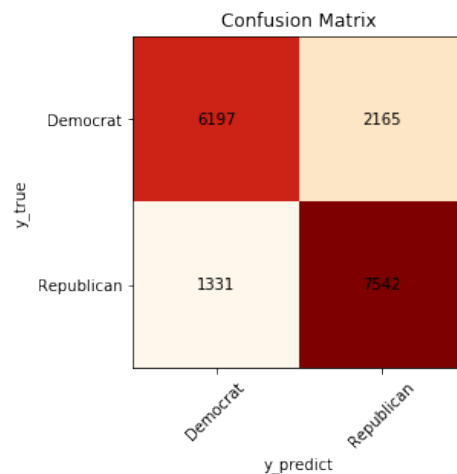


Figure 8. Confusion matrix

As you can see from the figure above, the number of Democrat tweets correctly predicted is 6,197, and the number incorrectly predicted is 2,165; The number of correctly predicted Republican tweets is 7,542, while the number of wrongly predicted Republican tweets is 1,331. It can be seen that the accuracy rate for Republican tweets is higher than for Democrat tweets. Overall, the model correctly classifies about 80% of tweets.

Step 7: Test the best model.

Finally, we tested the best model (multinomial naive Bayes) by entering the most recent tweet from a Republican Senator and a Democratic Senator. Of course, we can't test how well the model works on new data just from two tweets, but it's mostly just fun to enter text and see how the model behaves. To our delight, the model seems to do a good job with the Senators' tweets.

The most recent tweet from Ted Cruz (R, Texas) is "A whole lot of people in America have been sleeping on the danger that #China poses... @MichaelJKnowles

and I discuss in the latest episode of #Verdict." Our model correctly predicts that this tweet was written by a Republican.

The most recent tweet from Bernie Sanders (D, Vermont) is "The Walton family is worth \$165.7 billion. Do we really need to have families in America worth more than \$100 billion? When half of our people live paycheck to paycheck? Let us, together, have the courage to take on the greed of the corporate elite." Our model correctly predicted that this tweet was written by a Democrat.

7. Conclusion

Our text classification model was able to do a fairly decent job at picking up on the differences between the words that Democratic Representatives tended to use in their tweets and the words that Republican Representatives tended to use. However, we were only able to achieve 80% overall accuracy, which isn't outstanding. This may be due to a large number of Representatives falling somewhere in the middle of the political spectrum. Or perhaps the bag-of-words model doesn't do a great job in picking up the sentiment towards the topic. Perhaps there are some topics that both Republicans and Democrats would tweet about, but one would speak favorably and one would speak unfavorably, which the bag-of-words TFIDF model may have a hard time deciphering.

Overall, though, we're pretty satisfied with 80% accuracy. Going in to the project, we weren't sure if the model would be able to pick up on any differences at all. We found it very interesting that it seems that the language politicians use, and the topics they chose to discuss on social media can potentially paint a picture of where they fall on the political Left-Right spectrum.

8. Future work

The feature extraction method used in this paper, while fast and easy to execute, has some disadvantages. The TFIDF word vector cannot consider the position information of words, so it loses a lot of text feature information. Also, in theory, the naive Bayesian model has the smallest error rate compared with other classification methods, but this is not always the case because naive Bayesian models assume that the attributes are independent, which is often not true in practice.

In the future work, we can consider using a distributed word vector representation in the extraction of text features such as word2vec, Glove, or Fasttext. These are shallow neural network models used to extract word vector features, and they can better represent word order and semantics. For the classifier, we can consider introducing a deep learning model such as RNN or LSTM in order to achieve a better classification effect.

There could be other features we can include in a classification model as well, other than the raw text data. For example, which accounts do they follow, which accounts do they tend to reply to, and which accounts tend to reply to them? What websites do they tend to provide links to in their tweets? And when they provide links, does it tend to be videos, pictures, gifs, podcasts, articles, etc? What device was the tweet composed from? From what geographical location? What time of day? A lot of interesting work can be done in the future in feature extraction of twitter data.

9. References:

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